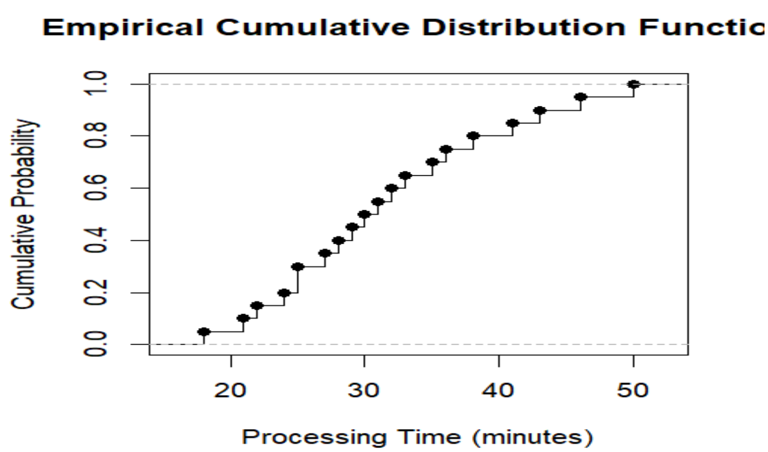
CO-5 AT2 – Visualization Analysis Report

Q1. Comparing Distributions – Histogram, Density and ECDF

R Code

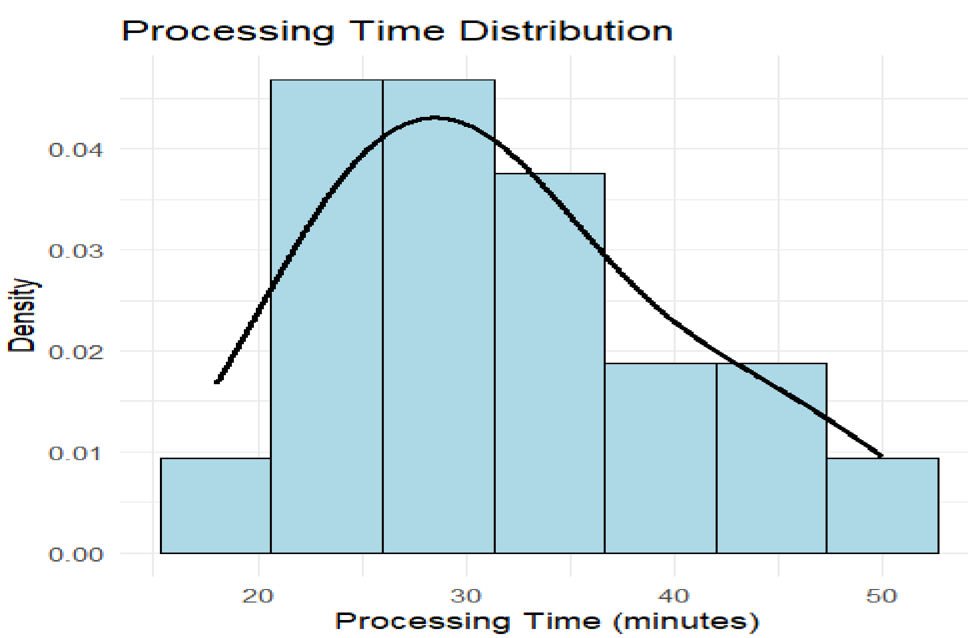
```r
# Data
Time <- c(18, 21, 22, 24, 25, 25, 27, 28, 29, 30,
        31, 32, 33, 35, 36, 38, 41, 43, 46, 50)

# Basic statistics
mean(Time)
median(Time)
sd(Time)

# -------------------------------
# Histogram
# -------------------------------
hist(Time,
    breaks = 7,
    main = "Histogram of Processing Times",
    xlab = "Processing Time (minutes)",
    ylab = "Frequency",
    col = "lightblue",
    border = "black")

# -------------------------------
# Density plot
# -------------------------------
```

```
plot(density(Time),
    main = "Density Plot of Processing Times",
    xlab = "Processing Time (minutes)",
    ylab = "Density",
    lwd = 2)

# -------------------------------
# ECDF
# -------------------------------
plot(ecdf(Time),
    main = "Empirical Cumulative Distribution Function",
    xlab = "Processing Time (minutes)",
    ylab = "Cumulative Probability",
    verticals = TRUE,
    do.points = TRUE)
```



Combined Histogram and Density Plot using ggplot2

```
library(ggplot2)


data <- data.frame(Time)


ggplot(data, aes(x = Time)) +
 geom_histogram(aes(y = after_stat(density)),
         bins = 7,
         fill = "lightblue",
         color = "black") +
 geom_density(linewidth = 1) +
 labs(title = "Processing Time Distribution",
    x = "Processing Time (minutes)",
    y = "Density") +
 theme_minimal()
```



Interpretation

The processing times range from **18 to 50 minutes**. The mean is approximately **31.2 minutes**, while the median is approximately **30.5 minutes**. The relatively larger values toward 41–50 minutes create a **slight right/positive skew**.

- **Histogram:** Clearly shows the frequency of observations in intervals. It communicates **center, spread, and overall shape** effectively.
- **Density plot:** Provides a smoothed representation of the distribution and makes the **shape and skewness** easier to identify.
- **ECDF:** Shows the cumulative proportion of observations. It is particularly useful for determining what percentage of tasks are completed below a particular time.

Comparison

Visualization	**Center**	**Spread**	**Skewness**	**Cumulative behavior**
Histogram	Good	Good	Good	Poor
Density	Good	Good	Excellent	Poor
ECDF	Moderate	Good	Moderate	**Excellent**

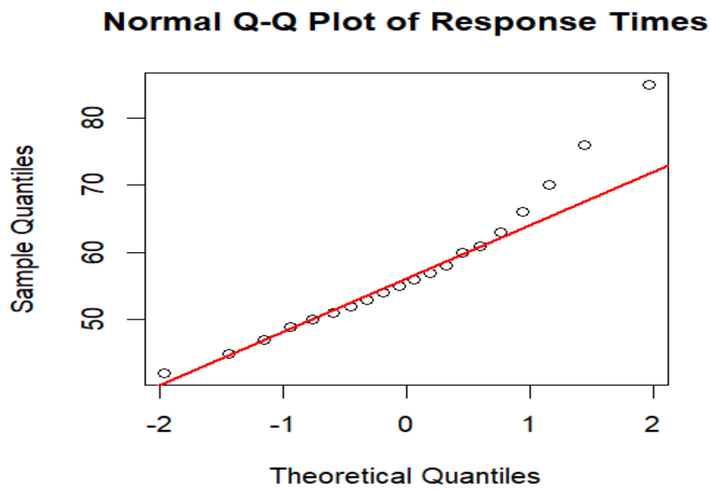
Conclusion: The histogram is most intuitive for frequency and spread, the density plot is best for understanding the smooth distributional shape and skewness, while the ECDF is best for cumulative probabilities and percentile-based interpretation.

Q2. Q-Q Plot and Normality Analysis

R Code

```
# Data

Response <- c(42, 45, 47, 49, 50, 51, 52, 53, 54, 55,

        56, 57, 58, 60, 61, 63, 66, 70, 76, 85)


# ------------------------------

# Histogram + Density
```

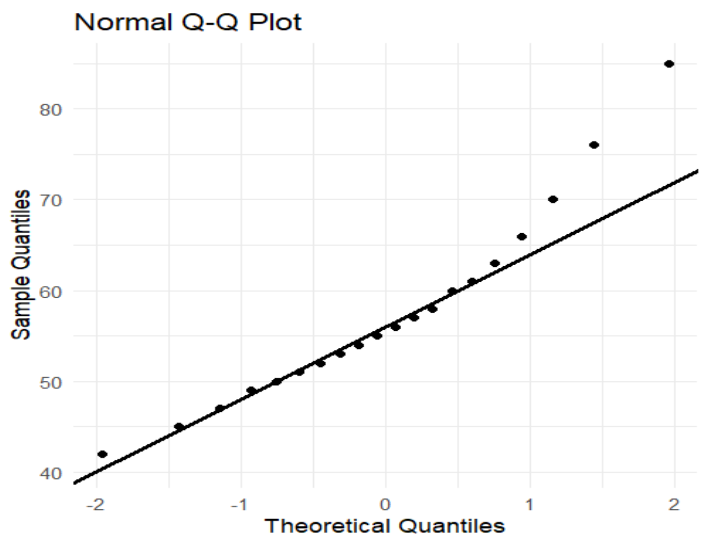
```
# -------------------------------
hist(Response,
    breaks = 8,
    probability = TRUE,
    main = "Histogram of Response Times",
    xlab = "Response Time (seconds)",
    ylab = "Density",
    col = "lightblue",
    border = "black")

lines(density(Response),
     lwd = 2)

# -------------------------------
# Normal Q-Q Plot
# -------------------------------
qqnorm(Response,
      main = "Normal Q-Q Plot of Response Times",
      xlab = "Theoretical Quantiles",
      ylab = "Sample Quantiles")

qqline(Response,
      col = "red",
      lwd = 2)
```



ggplot2 Version

```
library(ggplot2)


data <- data.frame(Response)


ggplot(data, aes(x = Response)) +
 geom_histogram(aes(y = after_stat(density)),
         bins = 8,
         fill = "lightblue",
         color = "black") +
 geom_density(linewidth = 1) +
 labs(title = "Response Time Distribution",
   x = "Response Time (seconds)",
   y = "Density") +
 theme_minimal()


ggplot(data, aes(sample = Response)) +
```

```
stat_qq() +
stat_qq_line(linewidth = 1) +
labs(title = "Normal Q-Q Plot",
    x = "Theoretical Quantiles",
    y = "Sample Quantiles") +
theme_minimal()
```



Interpretation

For normally distributed data, the points in the Q-Q plot should lie approximately along the reference line.

Here, most observations in the middle of the distribution follow the line reasonably well. However, the upper-tail observations, especially **76 and 85 seconds**, deviate substantially from the reference line.

The histogram also shows a **longer right tail**.

Therefore, the observations are **approximately normal in the central region but not perfectly normally distributed overall**. The upper-tail deviation suggests **positive skewness**, and 85 seconds can be considered a potential high-end outlier.

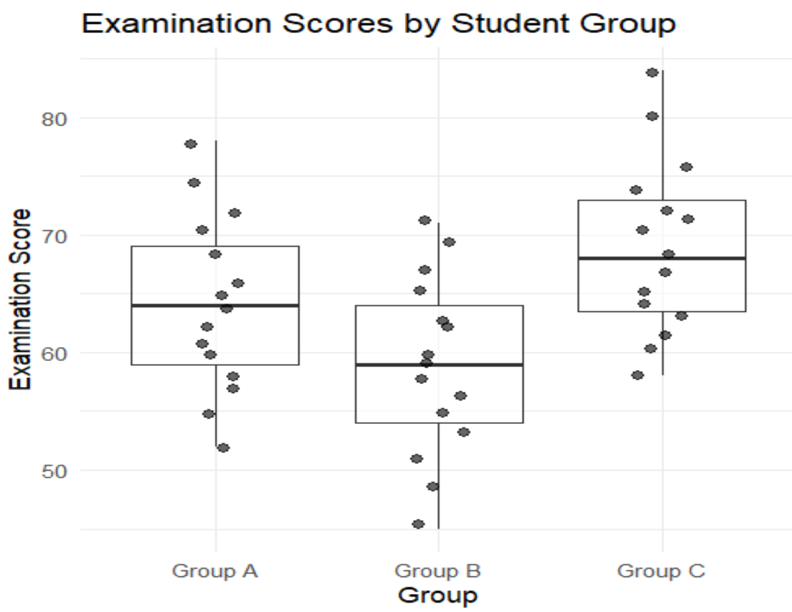
Conclusion: The Q-Q plot indicates that the normality assumption is questionable mainly because of the extreme upper-tail observations.

Q3. Visualizing Many Distributions at Once

R Code

```
library(ggplot2)


# Original data
Group_A <- c(52,55,57,58,60,61,62,64,65,66,
       68,70,72,74,78)


Group_B <- c(45,49,51,53,55,56,58,59,60,62,
       63,65,67,69,71)


Group_C <- c(58,60,61,63,64,65,67,68,70,71,
       72,74,76,80,84)


# Convert to long-format data
data <- data.frame(
  Group = rep(c("Group A", "Group B", "Group C"), each = 15),
  Score = c(Group_A, Group_B, Group_C)
)


# -------------------------------
# Boxplot + individual points
# -------------------------------
ggplot(data, aes(x = Group, y = Score)) +
  geom_boxplot(alpha = 0.6) +
  geom_jitter(width = 0.12,
```

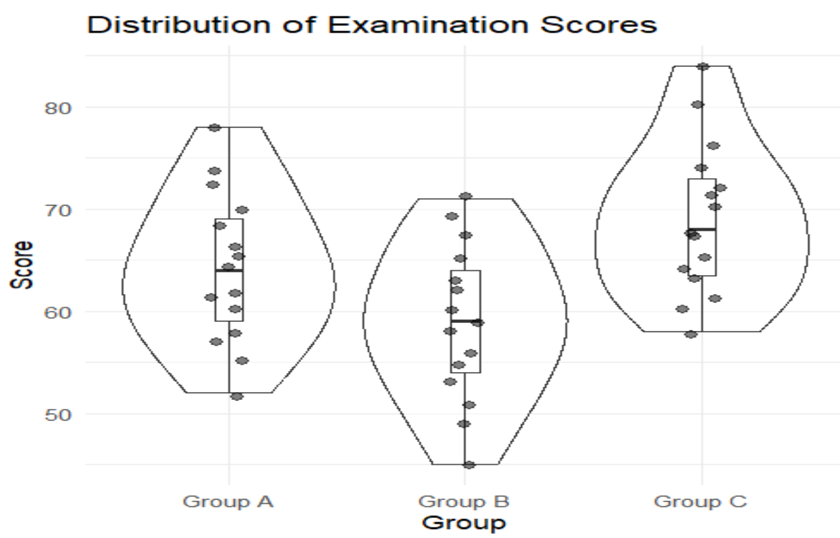
```
        alpha = 0.6,
        size = 2) +
  labs(title = "Examination Scores by Student Group",
     x = "Group",
     y = "Examination Score") +
  theme_minimal()
```



Violin Plot

```
ggplot(data, aes(x = Group, y = Score)) +
  geom_violin(alpha = 0.5) +
  geom_boxplot(width = 0.12) +
  geom_jitter(width = 0.08,
        alpha = 0.5,
        size = 2) +
  labs(title = "Distribution of Examination Scores",
     x = "Group",
     y = "Score") +
```

theme_minimal()



Interpretation

Approximate mean scores are:

Group	Mean Score
Group A	64.4
Group B	58.5
Group C	69.5

Group B has the lowest center, while **Group C** has the highest center.

- Group A has a moderate score distribution.
- Group B is shifted toward lower scores.
- Group C has the highest overall performance and extends to 84.
- There are no particularly extreme outliers based on the standard boxplot rule.

Why a combined plot is better

A grouped boxplot allows the three distributions to be compared **side-by-side using the same scale**. It immediately shows differences in median, spread, range, and possible outliers.

Separate plots require the viewer to mentally compare different graphics, making direct comparison less efficient.

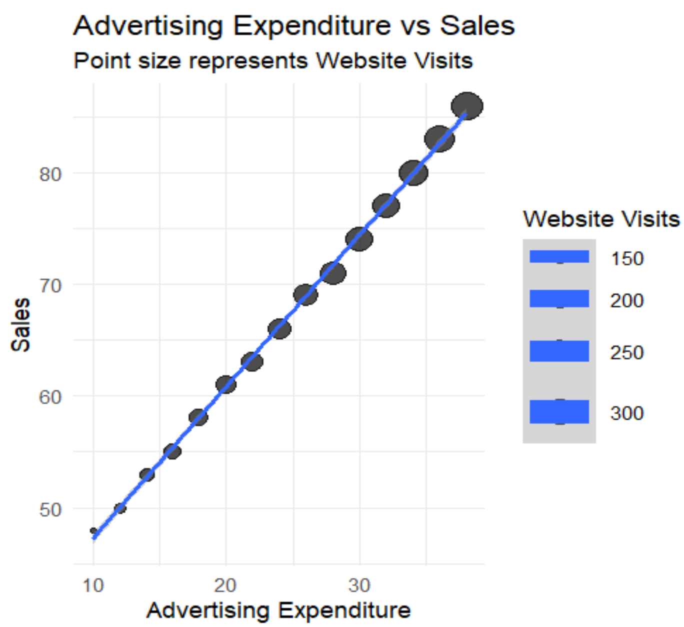
Conclusion: A boxplot with jittered observations is the most informative choice here because it combines a concise summary of each distribution with the actual student-level observations.

Q4. Relationships and Multivariate Visualization

R Code

```
Advertising <- c(10,12,14,16,18,20,22,24,26,28,30,32,34,36,38)


Visits <- c(120,135,150,165,180,195,210,225,245,260,

         275,290,305,320,340)


Sales <- c(48,50,53,55,58,61,63,66,69,71,

         74,77,80,83,86)


data <- data.frame(Advertising, Visits, Sales)


library(ggplot2)


# Scatterplot with website visits represented by point size

ggplot(data,

     aes(x = Advertising,

         y = Sales,

         size = Visits)) +

  geom_point(alpha = 0.7) +

  geom_smooth(method = "lm",
```

```
        se = TRUE) +
  labs(title = "Advertising Expenditure vs Sales",
       subtitle = "Point size represents Website Visits",
       x = "Advertising Expenditure",
       y = "Sales",
       size = "Website Visits") +
  theme_minimal()
```



Interpretation

There is a **very strong positive relationship** between advertising expenditure and sales. As advertising expenditure increases, sales consistently increase.

The fitted regression line clearly shows the upward trend.

Website visits also increase as advertising expenditure increases. Representing website visits through **point size** adds another dimension of information:

- X-axis → Advertising
- Y-axis → Sales
- Point size → Website visits

This helps show that higher advertising expenditure is associated with both increased website visits and increased sales.

However, using too many visual encodings can create **clutter**. For this relatively small dataset, point size is still easy to interpret.

Conclusion: The additional variable improves interpretation by providing context about website traffic, but the number of visual encodings should be limited so that the main advertising-sales relationship remains clear.

Q5. Misleading Visualization and Redesign

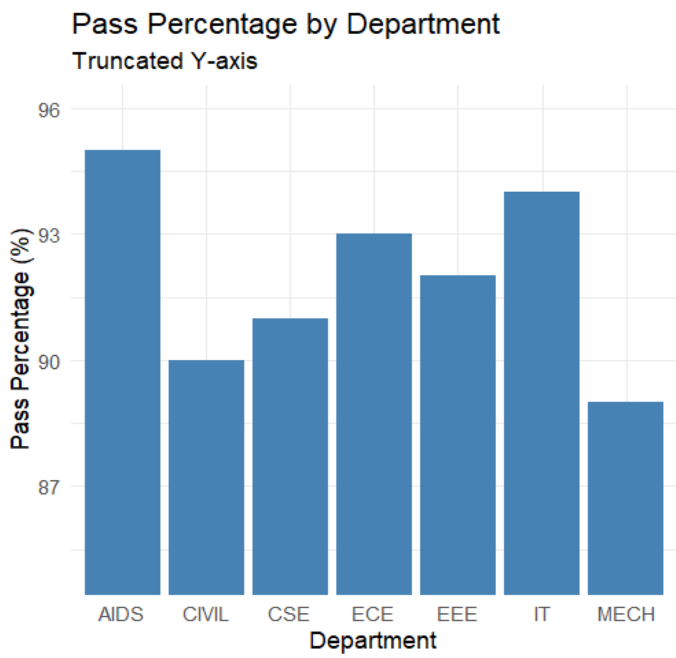
Data

```
Department <- c("CSE", "ECE", "EEE", "MECH",
        "CIVIL", "AIDS", "IT")


Pass_Percentage <- c(91, 93, 92, 89, 90, 95, 94)


data <- data.frame(Department,
         Pass_Percentage)
```

1. Misleading Visualization

```
library(ggplot2)


ggplot(data,
    aes(x = Department,
      y = Pass_Percentage)) +
 geom_col(fill = "steelblue") +
 coord_cartesian(ylim = c(85, 96)) +
 labs(title = "Pass Percentage by Department",
    subtitle = "Truncated Y-axis",
```

```r
    x = "Department",
    y = "Pass Percentage (%)") +
  theme_minimal()
```



This plot deliberately uses a truncated y-axis to demonstrate how visualization can exaggerate differences.

For example, the actual difference between the highest and lowest departments is:

$$95\% - 89\% = 6\%$$

Although the difference is only **6 percentage points**, the bars can appear dramatically different because the axis starts at 85 rather than 0.

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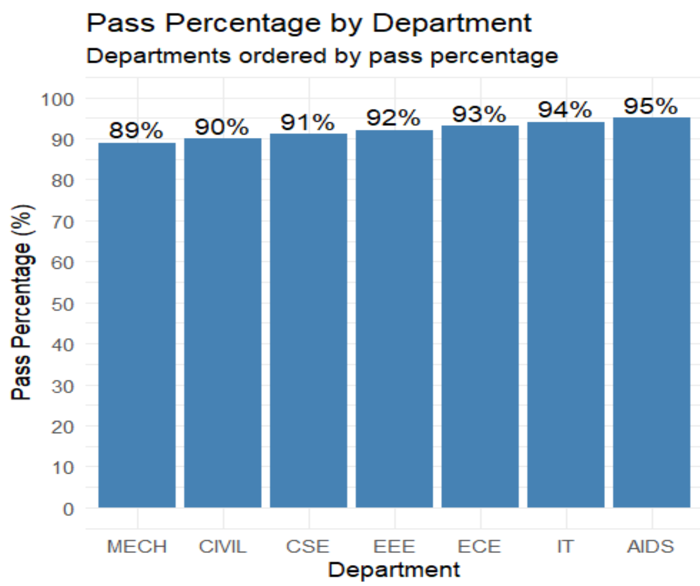
2. Redesigned Visualization

```r
# Order departments by pass percentage
data$Department <- reorder(
  data$Department,
  data$Pass_Percentage
)
```

```
ggplot(data,
    aes(x = Department,
      y = Pass_Percentage)) +
 geom_col(fill = "steelblue") +
 geom_text(
  aes(label = paste0(Pass_Percentage, "%")),
  vjust = -0.3,
  size = 4
 ) +
 scale_y_continuous(
  limits = c(0, 100),
  breaks = seq(0, 100, 10)
 ) +
 labs(title = "Pass Percentage by Department",
    subtitle = "Departments ordered by pass percentage",
    x = "Department",
    y = "Pass Percentage (%)") +
 theme_minimal()
```



Interpretation

The actual department ranking is:

Rank	Department	Pass %
1	MECH	89
2	CIVIL	90
3	CSE	91
4	EEE	92
5	ECE	93
6	IT	94
7	AIDS	95

All departments have relatively **high pass percentages**, ranging only from **89% to 95%.**

Comparison

Misleading version	Redesigned version
Y-axis begins at 85%	Y-axis begins at 0%

Misleading version	Redesigned version
Differences appear exaggerated	Differences shown proportionally
Departments are unordered	Departments are ordered
Exact values are less prominent	Exact percentages are labelled
Can create a false impression of large performance gaps	Gives an honest comparison

Visualization Pitfall

The major problem is the **truncated y-axis**. In a bar chart, the length of each bar represents magnitude, so starting the axis at a high value can make relatively small differences look substantial.

The redesigned chart uses a **zero baseline**, logical ordering, and direct labels. It therefore communicates that the departments are actually quite similar, despite the small ranking differences.

Conclusion: The redesigned visualization is more honest, accessible, and easier to interpret because it preserves proportional bar lengths and clearly communicates the actual 89–95% range.